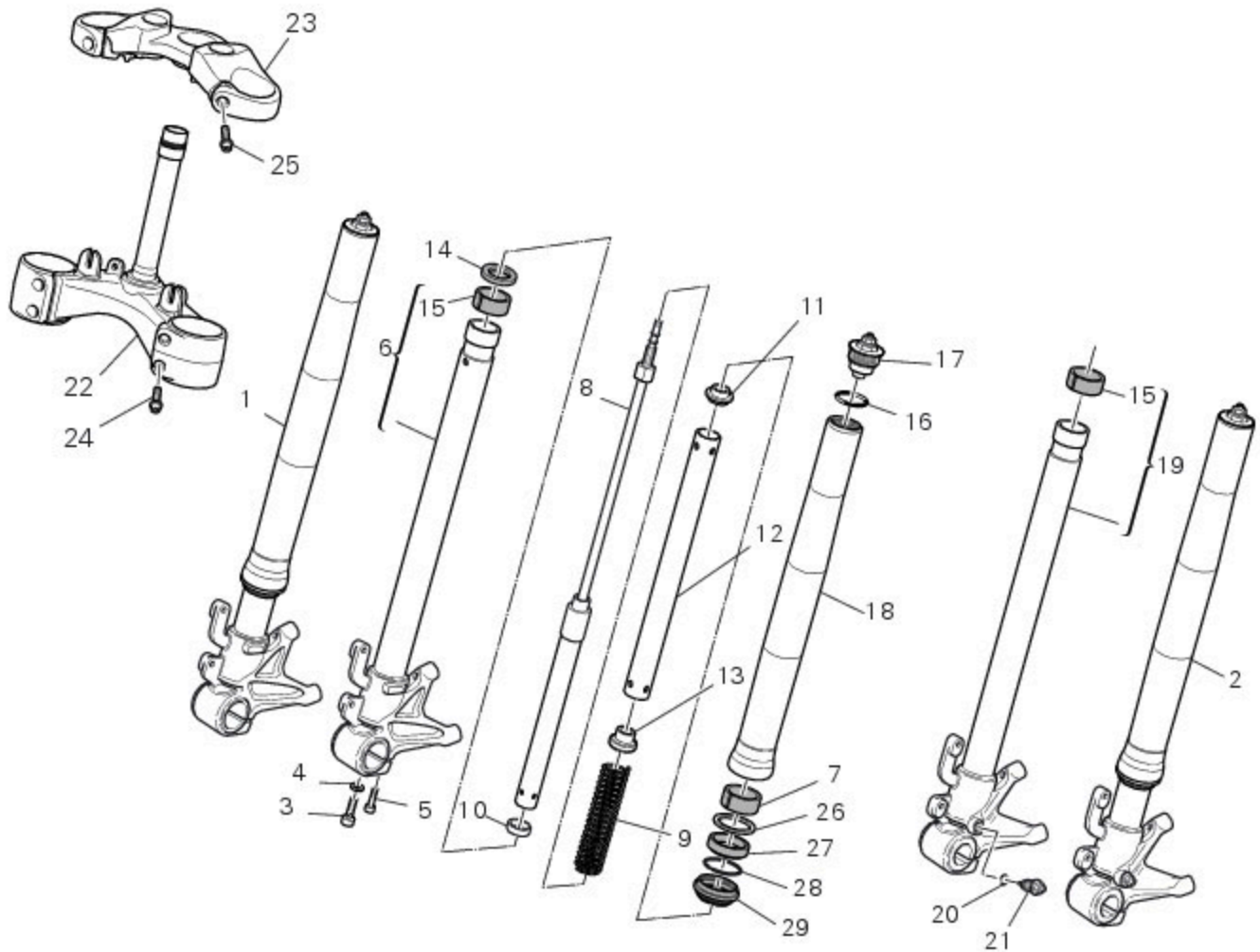


2.1 -Front fork (1100 ABS)



- 1 Right fork leg assembly
- 2 Left fork leg assembly
- 3 Banjo bolt
- 4 Special washer
- 5 Screw
- 6 Right inner tube assembly
- 7 Bush
- 8 Damper assembly
- 9 Spring
- 10 Centring bush
- 11 Joint
- 12 Preload tube
- 13 Spring retainer
- 14 Rubber washer
- 15 Bush
- 16 O-ring
- 17 Plug
- 18 Outer tube
- 19 Left inner tube assembly
- 20 O-ring
- 21 Adjuster screw
- 22 Bottom yoke
- 23 Steering head
- 24 Screw
- 25 Screw
- 26 Collar

27 Sealing ring
28 Circlip
29 Dust cap

 Spare parts catalogue

1100 ABS [FRONT FORK](#)
1100 ABS [HANDLEBAR AND CONTROLS](#)

Important

Bold reference numbers in this section identify parts not shown in the figures alongside the text, but which can be found in the exploded view diagram.

Refitting the front fork (1100 ABS)

Before removing the front forks, it is first necessary to remove the following parts:

Operation	Section reference
Draining the braking circuit	D 4, Draining the brake circuit
Removal of the front ABS sensor	G 9.2, Replacing the front phonic wheel sensor
Removal of the front brake callipers	G 3, Removal of the front brake system
Removal of the front mudguard	E 4, Removal of the front mudguard
Removal of the front wheel	G 1, Removal of the front wheel

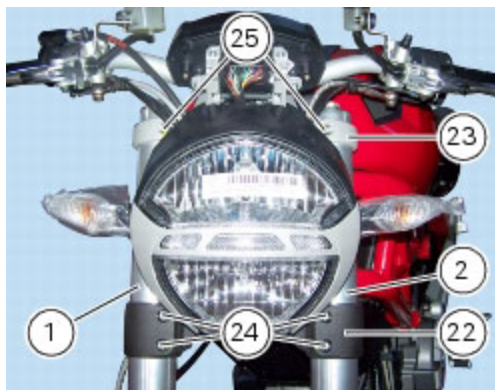
Important

Not all the components described in this chapter are available as individual replacement parts: however they are indicated for clearer explanation of the procedure.

Loosen the clamp screws (25) holding the fork legs to the steering head (23).

Loosen the clamp screws (24) securing the fork legs to the bottom yoke (22).

Withdraw the fork legs (2) and (1) downwards in order to carry out to all the necessary overhaul operations.



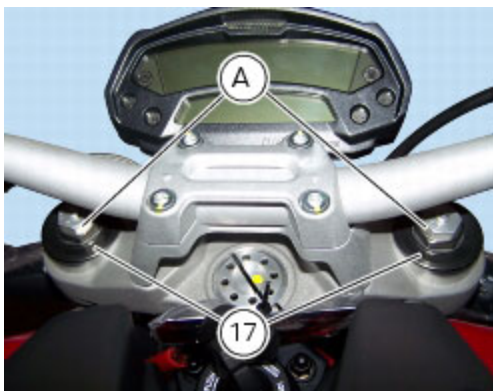
Overhauling the front fork (1100 ABS)

Note

It is advisable to loosen the top cap (17) when the fork leg is still fitted to the motorcycle.

The specific tools for the revision of the fork, are described in Sect. C 4, [Specific tools for the frame](#).

Loosen the spring preload adjuster (A) before unscrewing the plug (17).

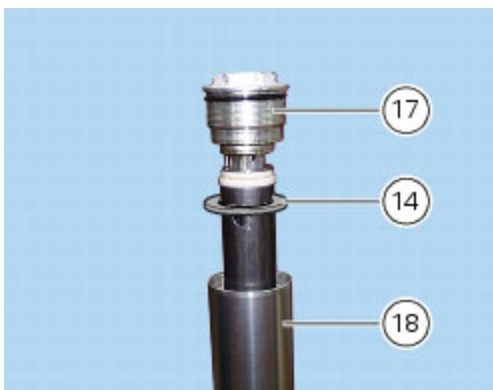


Disassembly of the front fork legs.

Unscrew the top cap (17).



Slide plug (17) off the stanchion tube (18), still mounted on the cartridge, and the rubber washer (14).

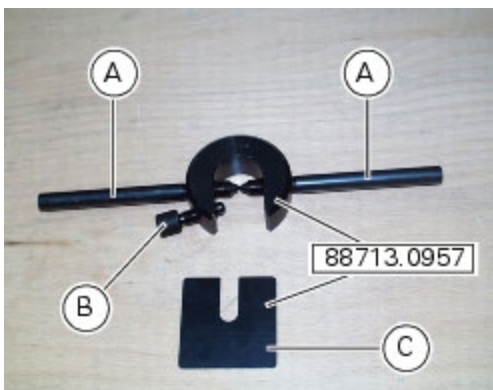


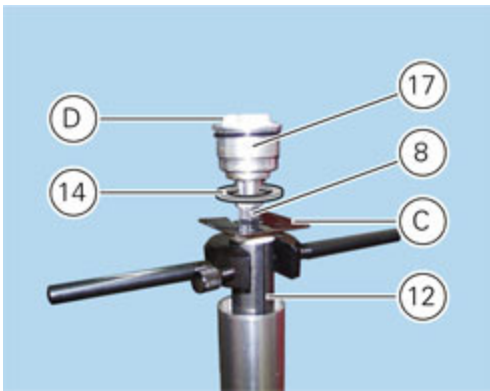
Install the special tool no. **88713.0957**, by inserting pins (A) in the holes of the preload tube and tightening screw (B).

Push the tool downwards to compress the spring and then fit the tool spacer (C) under the cartridge lock nut (8).

Apply a wrench to the flats (D) and counterhold the lock nut to loosen the top cap (17).

Remove the plug (17) and the rubber washer (14).

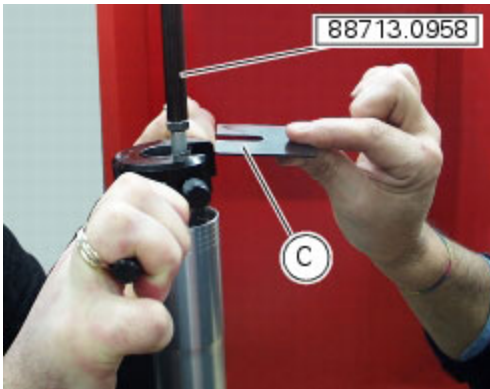




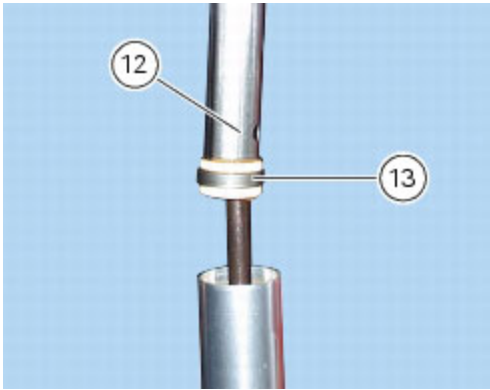
Install tool no. **88713.0958** screwing it onto cartridge rod.

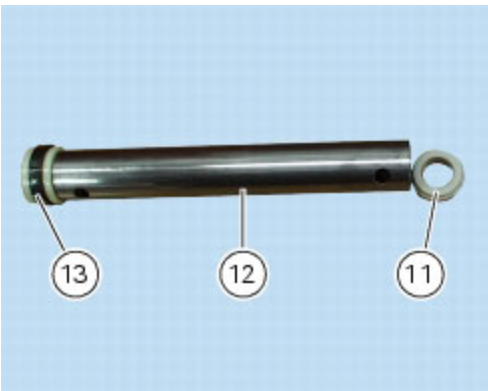


Push downwards to overcome the force of the spring and remove the spacer (C).



Remove tool no. **88713.0957** and slide out the preload tube (12) together with spring retainer (13) and joint (11).





Remove the spring (9) and pump the outer tube (18) and the damper (8) to expel the oil from the fork leg.

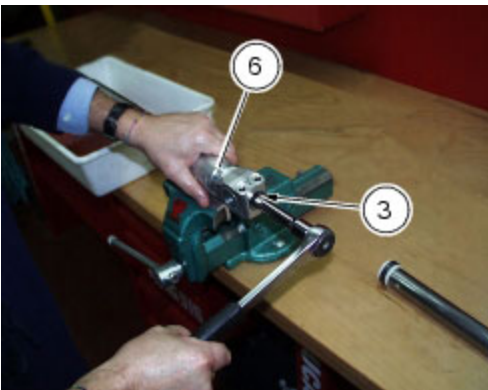


Warning

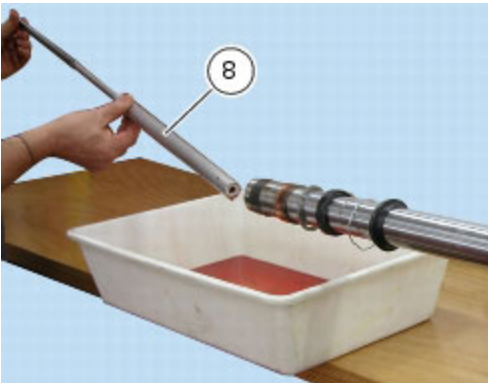
This movement generates pressure so that the oil will be rapidly expelled from the fork leg. Aim the jet of oil into a previously prepared container and avoid contact with fork oil.

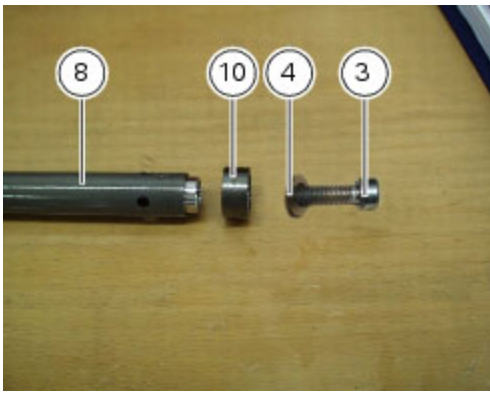


Clamp the inner tube (6) in a vice with soft-faced jaws. Unscrew the screw (3).

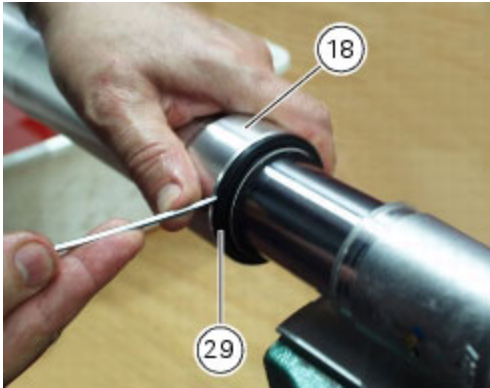


Remove the special screw (3), keep the special washer (4). Remove the complete damper assembly (8) and locating ring (10).

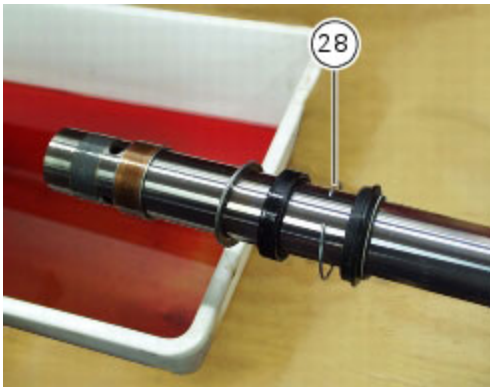




Slip the dust seal (29) off the outer tube (18), prising it off with a screwdriver.

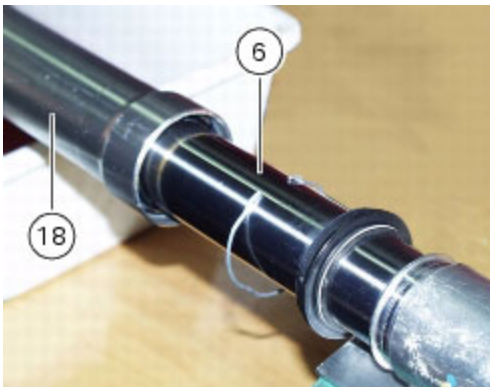


Remove the circlip (28).

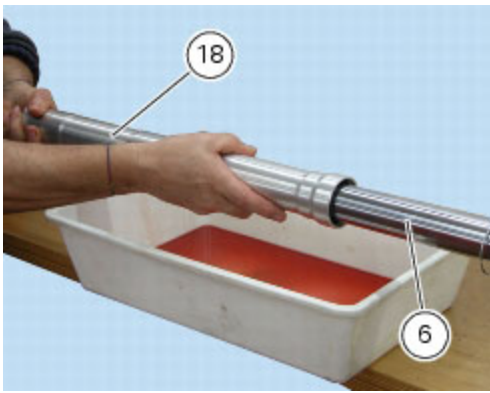


Important

Take care not to damage the outer tube (18) or inner tube (6).

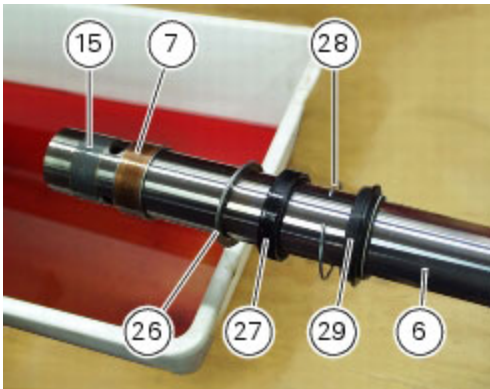


To separate the outer tube (18) from the inner tube (6), pull the inner tube sharply outwards repeatedly to gradually displace the bottom bush (7).
Insert the tip of a screwdriver in the slit in the bush (15) and open it up sufficiently to slide the bush off the inner tube.



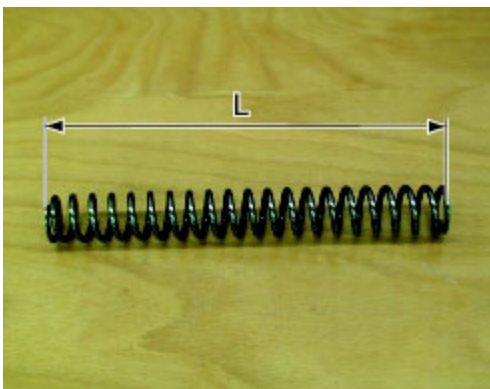
Remove the following parts from the inner tube (6):

- bush (7);
- collar (26);
- sealing ring (27);
- circlip (28);
- dust seal (29).



Inspection of the front fork (1100 ABS).

Place the spring on a flat surface and measure the free length (L).



Service limit:
298.5 mm.

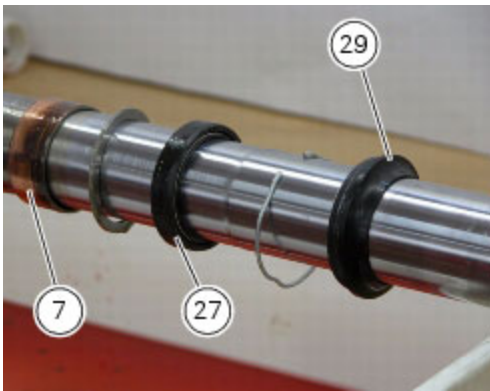
Renew the spring if the length is not within the specified limit.

Inspect the outer surfaces of both fork inner tubes and the internal surfaces of both outer tubes. They must be free from scoring, steps, or dents.

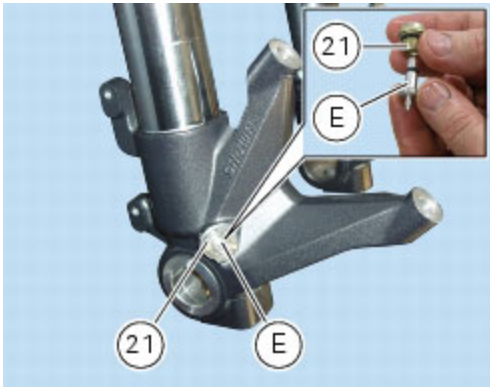
Check that the outer tubes are straight:

maximum allowed error **0.10** mm.

Whenever the inner tube is removed from the outer tube it is good practice to renew the bushes (15) and (7) as well as the seals (27) and dust seal (29).



Loosen and remove the plug (21) located on fork bottom end.
Remove the taper needle (E).



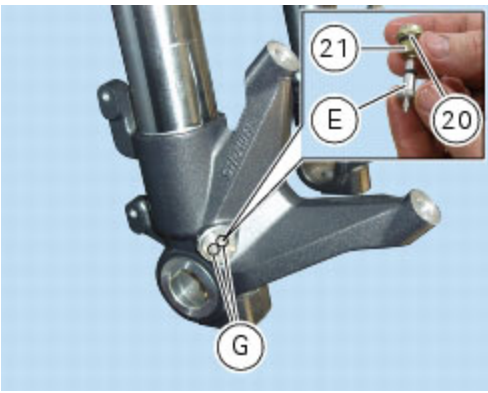
Reassembly of the front fork (1100 ABS)

Clamp the inner tube in a vice.

Protect the end of the stanchion tube with tape.

Fit the duly greased O-ring (20) onto plug (21).

Upon reassembly, screw the taper needle (E) so that it seats against the fork bottom end, then tighten plug (21) until reference marks (G) are perfectly aligned.

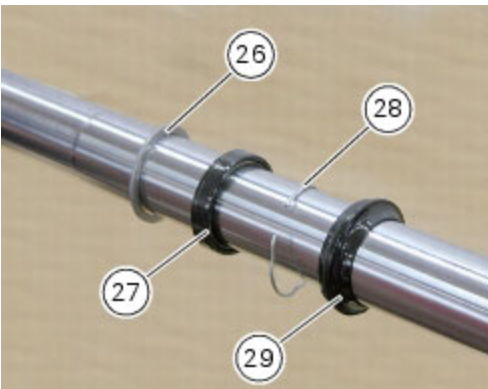


Important

Before installation, smear the sliding edges of oil sealing ring (27) with fork oil or oil seal grease.

Fit the following components to the inner tube:

- dust seal (29);
- circlip (28);
- sealing ring (27);
- collar (26).



Important

Install the sealing ring with the marked surface facing the dust seal.

Fit the following components:

- bush (7);
- bush (15).



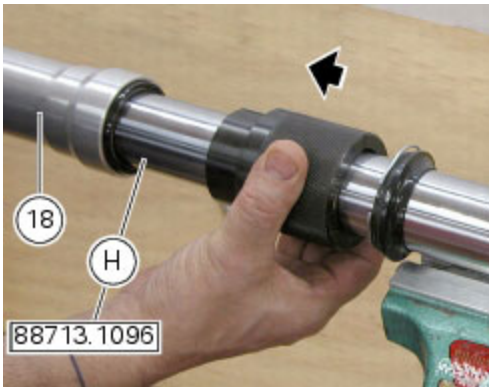
Remove the adhesive tape from slider end.



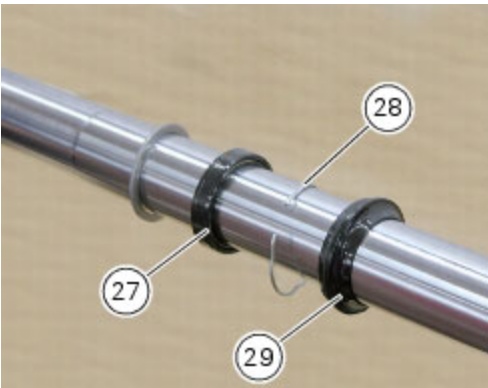
Note

Before assembling the two parts, lubricate the sliding surfaces of the bushes with fork oil.

Install the outer sleeve (18) and push the guide bush (7) and the cap (26) into the outer sleeve using the sealing ring tool (H, part no. **88713.1096**).



Drive the oil sealing ring (27) into the outer tube using the same tool. Fit the circlip (28) and the dust seal (29).

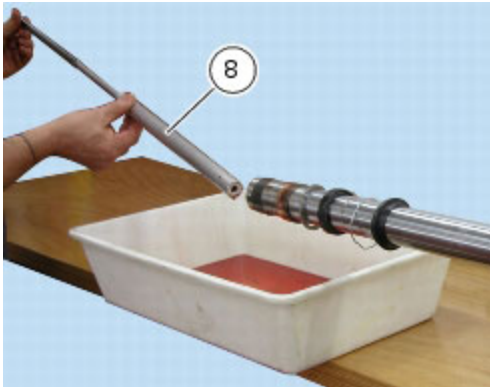
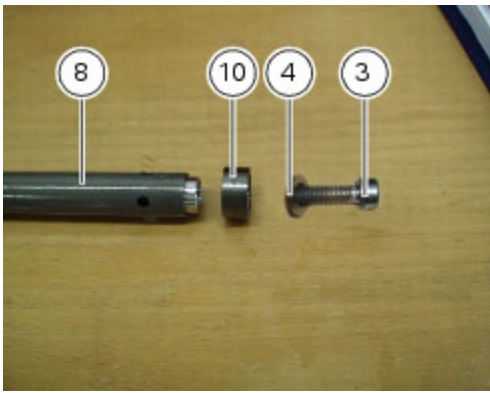


Important

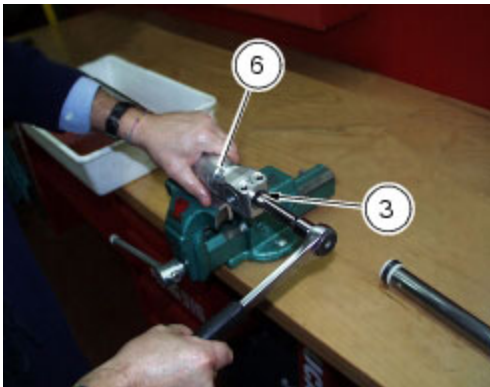
The outer sleeve must slide freely along the slider. Only support the sliders and the outer sleeve with your hands so not to damage the sealing rings and the pilot bushes.

Insert the locating bush (10).

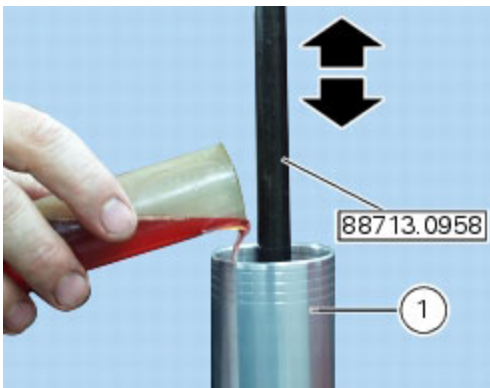
Fit the complete damper cartridge (8) on the slider (6).



Fit the seal (4) and tighten the adjuster screw (3).



Hold the left fork leg assembly (1) in a vertical position and fill with fork oil, using 524 ± 2.5 cc of the specific oil. Install tool part no. **88713.0958** to cartridge. Pump to force the oil to fill internal volume completely.



Check the quantity of oil measuring the distance of the level from the end of the outer tube (18). This distance should be **90 mm**. Clean off any oil from the spring (9) joints (11), (13) and the tube (12) before refitting.

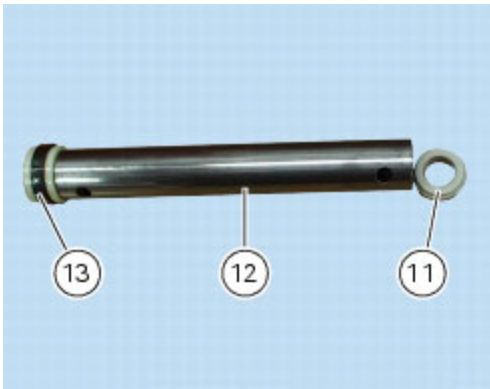
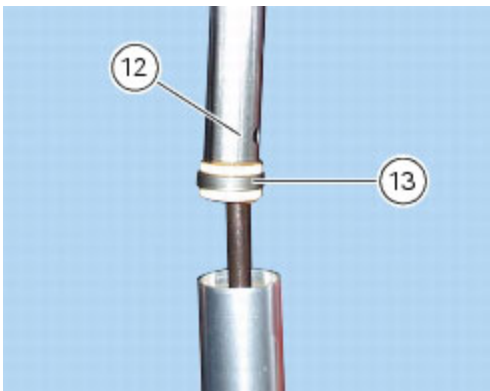


Note

Install the spring (9), ensuring that the end with tighter coils is towards the bottom of the fork.

Fit the following components:

- spring (9);
- spring retainer (13);
- preload tube (12);
- joint (11).

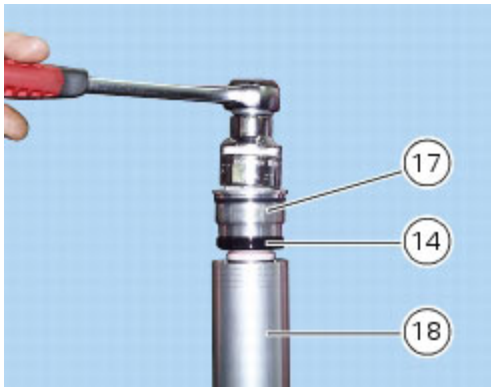
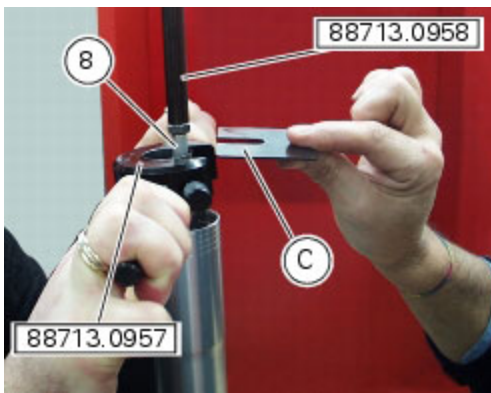


Install tool no. **88713.0957**, i.e. the one used for removing the preload tube (**12**). Press tool down and set spacer (C) under the cartridge (8) lock nut. Remove service tool no. **88713.0958**. Fit the rubber washer (14) and tighten top plug (17) on the damper cartridge (8). Hold cartridge lock nut with a hexagon wrench then tighten the top plug to **20±2.5 Nm**. Remove spacer (C) by pushing tool (part no. **88713.0957**) downward.

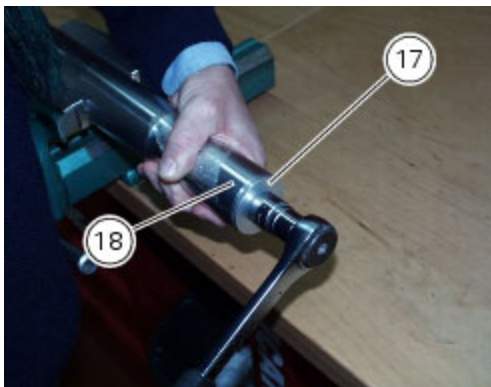


Warning

Plug (17) adjuster (A) shall be fully home against damper cartridge (8) thread.

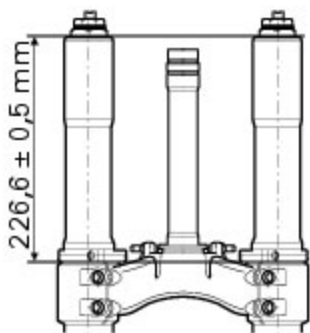


Lift the outer sleeve (18) and tighten the top plug (17) to a torque of **34 ± 5** Nm. Set the adjusters according to riding requirements.



Refit the front fork (1100 ABS)

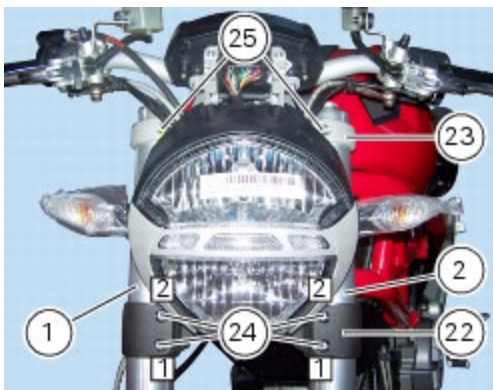
Refit the fork legs, positioning them at the height shown in the figure relative to the upper surface of the bottom yoke.



Warning

The difference in height between the two legs must not exceed 0.1 mm.

Position the fork legs, then tighten the screws (24) and (25) to the specified torque (Sect. C 3, [Frame torque settings](#)) following a 1-2-1 sequence for the screws (24).



Important

If the clamp screws (24) and (25) were removed on disassembly, smear the threads with the specified grease before tightening.



Warning

Do not use the motorcycle without the front mudguard fitted to avoid the risk of the brake pipes fouling the wheel on braking.

Refit any parts removed from the frame:

Operation	Section reference
Refitting the front wheel	G 1, Refitting the front wheel
Refitting the front mudguard	E 4, Refitting the front mudguard
Refitting the front brake callipers	G 3, Refitting the front brake system
Refitting the front ABS sensor	G 9.2, Replacing the front phonic wheel sensor
Filling the braking system	D 4, Filling the brake system with fluid